Oryna Ivashtenko

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Weizmann Institute of Science, Rehovot, Israel +972 58 637 8296

Research Interests Statistically calibrated detection, parameter estimation, and population inference for weak astrophysical signals in complex noise, with applications to exoplanets, stars, and compact objects.

Education

2022–2027 **Ph.D. in Physics, Weizmann Institute of Science**
Expected 2027. Advisor: Barak Zackay
Thesis: Detection methods and population studies of extrasolar planets.

2019–2022 **M.Sc. in Physics, Weizmann Institute of Science**
GPA 92.5
Advisor: Barak Zackay. Thesis: Looking for "blue" planets orbiting "yellow" stars.

2015–2019 **B.Sc. in Applied Physics, V. N. Karazin Kharkiv National University** (Ukraine)
With honors, GPA 96.9

Publications

2025 **O. Ivashtenko** and B. Zackay. An Independent Search for Small Long-period Planets in Kepler Data I: Detection Pipeline. AJ, 170, 303.

2026 **O. Ivashtenko** and B. Zackay. Marginalized planet-to-star radius ratio posteriors for low-SNR transiting planets via dimensionality reduction. arXiv:2609.23159.

2022 S. Shahaf, B. Zackay, T. Mazeh, S. Faigler, and **O. Ivashtenko**. fBLS – a fast-folding BLS algorithm. MNRAS 512, 2732–2746.

2025 J. Mushkin, J. Roulet, B. Zackay, T. Venumadhav, **O. Ivashtenko**, D. Wadekar, A. K. Mehta, and M. Zaldarriaga. Sampler-free gravitational wave inference using matrix multiplication. Phys. Rev. D 112, 104025.

2026 Z. Zhou, D. Wadekar, J. Roulet, **O. Ivashtenko**, T. Venumadhav, T. Islam, A. K. Mehta, J. Mushkin, M. Ho-Yeuk Cheung, B. Zackay, and M. Zaldarriaga. Searching for precessing binary systems with mode-by-mode filtering and marginalization. arXiv:2603.05784.

Awards and Fellowships

2024 FGS Dean Award of Excellence in Physics/Chemistry/Math & CS-oriented Research for Outstanding International Doctoral Students

2023 Ph.D. Scholarship in Data Science, Weizmann Data Science Research Center, supported by the Israeli Council for Higher Education

2022 The Suzan Sapir Memorial Prize for M.Sc. students

2019 Prize of Excellence, Andre Deloro Research School

2017–2019 Scholarship of the Ukrainian Parliament

Talks and Research Visits

2026 SPI-MAX Seminar, University of Oxford (Oxford, UK) – Invited.

2026 IvS Seminar, KU Leuven (Leuven, Belgium)

2026 ARCO seminar, Open University (Raanana, Israel) – Invited.

2026 Origins Seminar, the University of Arizona (Tucson, AZ)

2025 Astrolunch, Hebrew University (Jerusalem, Israel) – Invited

2025 Exocoffee Seminar, MPIA (Heidelberg, Germany)

2025 Seminar, DLR (Berlin, Germany)

2025 TESS Science Talks, MIT (Cambridge, MA) – Invited

2025 Tea Seminar, California Institute of Technology (Pasadena, CA)

2025 Lunch talk, UCSC (Santa Cruz, CA)

2025 Seminar, University of Washington (Seattle, WA)

2025 Seminar, SETI Institute (Mountain View, CA)

2024 Astrocoffee seminar and visiting student, Institute for Advanced Study (Princeton, NJ)

2023 Seminar, ETH Zürich (Zürich, Switzerland)

2023 Seminar, Penn State University (State College, PA)

2023 Seminar, Princeton University (Princeton, NJ)

2023 Seminar and visitor, Geneva Observatory (Geneva, Switzerland) – Invited

Conference presentations

2026 Exoplanets 6, Porto, Portugal – Poster

2026 247th meeting of the American Astronomical Society, National Harbor, MD – Talk

2025 70th Annual Meeting of the Israel Physics Society, Technion, Haifa, Israel – Talk

2025 Know Thy Star Know Thy Planet 2, Caltech, Pasadena, CA – Poster

2025 245th meeting of the American Astronomical Society, National Harbor, MD – Talk

2024 243rd meeting of the American Astronomical Society, New Orleans, LA – Talk

Teaching and Mentoring

2026 Mentor, undergraduate research project, Weizmann Institute of Science.

2026 Tutoring M.Sc. thesis student, Weizmann Institute of Science.

2025 Python tutor, Weizmann Medical School.

2021–2025 Teaching assistant, Statistics, Algorithms and Experimental Design, Weizmann Institute of Science.

2023–2024 Tutor, AI-Hub Data Science Project, Weizmann Institute of Science.

2021 Tutoring an experimental project M.Sc. student.

2016–2019 Instructor, Summer School in Mathematics, V. N. Karazin Kharkiv National University.

Academic Service and Outreach

2021–2024 Astronomy outreach lectures for Israeli and Ukrainian communities.

2016–2018, 2023 Trainer, jury member, and team leader, Ukrainian Young Physicists' Tournament.

2019 Team leader, Team Ukraine, 11th International Physicists' Tournament, Lausanne.

Technical Expertise

Methods Statistical detection theory, Bayesian inference, time-series analysis, Monte Carlo methods, injection–recovery calibration, empirical background estimation, large-scale numerical inference.

Computing Python, high-performance and distributed computing (CPU).